Homework #4 Solution

Problem 8.2 (a)

$$u(t) = u_s(t) - u_s(t-1) - u_s(t-2) + u_s(t-3)$$

$u_s(t)$ is a step input where $u_s(t) = \begin{Bmatrix} 0 & t \le 0 \\ 1 & t > 0 \end{Bmatrix}$

(b)

$$u(t) = u_r(t-1) - 2u_r(t-2) + u_r(t-3)$$

Problem 8.9 (c)

Find y_h

$$y_h = ce^{\lambda t}$$

$$\dot{y_h} = c\lambda e^{\lambda t}$$

$$\ddot{y_h} = c\lambda^2 e^{\lambda t}$$

$$c\lambda^2 e^{\lambda t} + 6c\lambda e^{\lambda t} + 9ce^{\lambda t} = 0$$

$$\lambda^2 + 6\lambda + 9 = 0$$

$$\lambda_{1,2} = -3$$

$$y_h(t) = c_1 e^{-3t} + c_2 te^{-3t}$$

$$y_h(0) = 1 => c_1 = 1$$

$$\dot{y_h}(0) = 0 => -3c_1 + c_2 = 0 => c_2 = 3$$

$$y_h = e^{-3t} + 3te^{-3t}$$

Problem 8.11

$$\text{coswt} = \frac{1}{2}\left(e^{jwt} + e^{-jwt}\right)$$

$$e^{at} coswt = \frac{1}{2} e^{at+jwt} + \frac{1}{2} e^{at-jwt}$$

Here, a = -2, w = 4

$$u = 6e^{-2t}\cos(4t) = \frac{1}{2}6\left(e^{-2t+4jt} + e^{-2t-4jt}\right)$$

$$= 3e^{-2t+4jt} + 3e^{-2t-4jt}$$

Assume

$$y_p = K_1 e^{-2t+4jt} + K_2 e^{-2t-4jt}$$

$$\dot{y_p} = (-2+4j)K_1 e^{-2t+4jt} + (-2-4j)K_2 e^{-2t-4jt}$$

Substitute

$$\dot{y_p} + y_p = (-2K_1 + 4jK_1 + K_1)e^{-2t+4jt} + (K_2 - 2K_2 - 4jK_2)e^{-2t-4jt}$$

$$= 3(e^{-2t+4jt} + e^{-2t-4jt)}$$

$$\begin{Bmatrix} -2K_1 + 4jK_1 + K_1 = 3 \\ K_2 - 2K_2 - 4jK_2 = 3 \end{Bmatrix}$$

$$\begin{Bmatrix} K_1 = \dfrac{3}{4j-1} \\ K_2 = -\dfrac{3}{4j+1} \end{Bmatrix}$$

$$y_p = \frac{3}{4j-1}e^{-2t+4jt} - \frac{3}{4j+1}e^{-2t-4jt}$$

$$= \frac{12j+3}{(4j-1)(4j+1)}e^{-2t}(cos4t + jsin4t) - \frac{12j-3}{(4j-1)(4j+1)}e^{-2t}(cos4t - jsin4t)$$

$$= e^{-2t}(-\frac{12j+3}{17}cos4t - \frac{-12+3j}{17}sin4t + \frac{12j-3}{17}cos4t + \frac{12+3j}{17}sin4t$$

$$= e^{-2t}\left(\frac{24}{17}sin4t - \frac{6}{17}cos4t\right)$$

Problem 8.12

$$\frac{d^2y}{dt^2} + 9\frac{dy}{dt} + 20y = u(t)$$

$$u(t) = 5sin2t$$

Assume

$$y_p = K_1 cos2t + K_2 sin2t$$

$$\dot{y_p} = -2K_1 sin2t + 2K_2 cos2t$$

$$\ddot{y_p} = -4K_1 cos2t - 4K_2 sin2t$$

$$-4K_1 cos2t - 4K_2 sin2t - 18K_1 sin2t + 18K_2 cos2t + 20K_1 cos2t + 20K_2 sin2t = 5sin2t$$

$$\begin{Bmatrix} -4K_1 + 18K_2 + 20K_1 = 0 \\ -4K_2 - 18K_1 + 20K_2 = 5 \end{Bmatrix}$$

$$\begin{Bmatrix} 18K_2 + 16K_1 = 0 \\ -18K_1 + 16K_2 = 5 \end{Bmatrix} => K_2 = -\frac{8}{9}K_1$$

$$-18K_1 - \frac{16 * 8}{9}K_1 = 5$$

$$K_1 = -\frac{9}{58}, \qquad K_2 = \frac{4}{29}$$

Problem 8.18 (b)

$$y_h = ce^{\lambda t}$$

$$CE: \lambda^2 + \lambda - a = 0$$

$$\lambda = \frac{-1 \pm \sqrt{1 + 4a}}{2}$$

$$y_h = C_1 e^{-1 + \frac{\sqrt{1+4a}}{2}t} + C_2 e^{-1 - \frac{\sqrt{1+4a}}{2}t}$$

Stable:

$$\frac{-1 + \sqrt{1 + 4a}}{2} < 0$$

$$-1 + \sqrt{1 + 4a} < 0$$

$$\sqrt{1 + 4a} < 1$$

$$1 + 4a < 1$$

$$4a < 0$$

$$\underline{\underline{a < 0}}$$

$$\frac{-1 - \sqrt{1 + 4a}}{2} < 0$$

$$-1 - \sqrt{1 + 4a} < 0$$

$$-\sqrt{1 + 4a} < 1$$

$$\underline{satisfy}$$

$$Marginally\ stable: a = 0$$

$$Unstable: a > 0$$

8.18(b)

$$y_h = ce^{\lambda t}$$

$$\frac{d^2y_h}{dt^2} + \frac{dy_h}{dt} - ay_h = 0$$

$$\lambda^2 + \lambda - a = 0$$

$$\lambda_{1,2} = \frac{-1 \pm \sqrt{1+4a}}{2}$$ For stable system, $Re(\lambda) \le 0$

condition 1: $1+4a \le 0$

$a \le -\frac{1}{4}$ stable because $Re(\lambda_{1,2}) = -\frac{1}{2} < 0$

condition 2: $1+4a > 0$ $a > -\frac{1}{4}$

$$\lambda_1 = \frac{-1+\sqrt{1+4a}}{2} < 0$$

$$\lambda_2 = \frac{-1-\sqrt{1+4a}}{2} < 0$$

satisfy λ_1 will automatically satisfy λ_2

$$\lambda_1 = \frac{-1+\sqrt{1+4a}}{2} < 0$$

$$\sqrt{1+4a} < 1$$

$$1+4a < 1$$

$$4a < 0$$

$$a < 0$$

when $-\frac{1}{4} < a < 0$, system stable.

combine condition 1 and condition 2 results, to reach stability, $a < 0$